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DEVELOPMENT OF A FLAMELET PROGRESS VARIABLE APPROACH FOR USE IN A FULLY CONSERVATIVE DISCONTINUOUS GALERKIN NAVIER-STOKES SOLVER

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ABSTRACT

The flamelet progress variable (FPV) approach is advantageous in its abilities to significantly reduce the number of partial differential equations to be solved and the numerical stiffness associated with chemically reacting Navier-Stokes equations. In diffusion flame scenarios, two transport equations, one for the conserved scalar of mixture fraction (Z), used to define the level of mixing between fuel and oxidizer and the other, the combustion progress variable (C_{pv}), used as an indicator of combustion progression from the pure mixing case to that of equilibrium combustion chemistry are used to describe the combustion environment. An improved method for determining the species and weights used to defined C_{pv} is presented. A downhill simplex method is used to minimize the cost function which is a combination of Heaviside functions used to ensure monotonicity, flame ordering, minimized gradients, and a normalized C_{pv} bounded between 0 and 1. One-dimensional flames, spanning from low to high strain are tabulated within the flamelet generated manifold (FGM) for species and C_{pv} source term lookup during run time. The tabulated species are used to determine the enthalpy diffusive flux in the total energy conservation equation, as well as species weighted thermodynamic and transport properties. The FPV approach is coupled with a full conservative Discontinuous Galerkin (DG) flow solver and simulations of a backward facing step combustion problem is explored. This approach has the ability to significantly decrease computational cost of chemically reacting flows while still maintaining details associated with finite-rate chemistry.

INTRODUCTION

The flamelet progress variable (FPV) and flamelet generated manifold (FGM) [1, 2, 3] method has been used to describe premixed flames [4, 5] and non-premixed flames [6, 3] with extensions to multi-dimensional turbulent combustion environments. Knudsen and Pitsch [7] have studied the combination of premixed and non-premixed flamelets to assess their mutual application to describe partially-premixed combustion environments. Wu et al. [8] introduced the Pareto-efficient combustion model for predicting and modeling multi-regime flames. Vreman et al. [9] and Ihme et al. [10] have applied the FGM method to complex turbulent combustion simulations of the experimental set of Sandia D and E flames developed by Barlow et al. [11]. The FGM approach has also been extended to modeling dilute spray combustion including *n*-heptane [12], α-methylnaphthalene/*n*-decane [13], *n*-dodecane [14], acetone sprays [15], and Jet-A fuel [16]. This approach has also been successfully adapted for multiphase combustion problems with reacting interfaces [17] and extended for turbulent spray flames, accounting for the effects of varying combustion modes from individual droplet combustion to group combustion [18]. Solving the complete reacting Navier-Stokes equations with hundreds of species remains intractable with multiple researchers attempting to eliminate the addition partial differential equations (PDEs) introduced for each species and the numerical stiffness associated with the reaction source terms, these include new approaches to tabulation [19] and reducing assumptions like unity Lewis number ($Le = 1$) [20].

Other limiting assumptions include using the table to fully resolve the chemical state at the subgrid scale in the context of large eddy simulation (LES) approaches. This includes the mixture weighted thermodynamic values which are a function of the combustion temperature, all of which are tabulated. This requires the conservation of energy to be completely neglected through the use of the lookup table or to be severely limited by relying so heavily on the table to determine the temperature of the gas, neglecting compressibility and dissipative effects. The advantages of using tabulated chemistry, as stated earlier, is primarily to eliminate the number PDEs to be solved and the numerical stiffness, while maintaining the effects of detailed combustion chemistry capable of predicting dynamic events like ignition, extinction, and liftoff. To this end, we propose maintaining the total energy conservation equation and using the chemical state, as defined by Z and C_{pv}, and the total energy to define the temperature of the gas, which in turn is used determine the thermodynamic properties. Therefore, the only dependent variables needed from the tables are mass fractions (Y_i's) and the progress variable source term ($\dot{S}'''_{C_{pv}}$) to describe the time rate of change of the chemical state information.

Two critical pieces are needed to implement the FPV approach in the context of a Discontinuous Galerkin (DG) solver described in this study. First, is the definition of the FGM data derived from a progression of $1D$ flame solutions, with particular focus on the definition of the progress variable (C_{pv}) and its implications. Secondly, the conservation equations for Z and C_{pv}, as well as for mass, momentum, and energy need to be clearly defined in the context of a fully compressible and conservative DG approach. The remainder of this study is organized as follows. The governing equations of the $1D$ flame equations are described along with assumptions needed for their tabulation. Then, the mathematical descriptions of C_{pv} and cost function approach to determine species weights is given in detail with a follow-on discussion of how the tabulation data is obtained in its final form. The next section describes the DG numerical model used to solve the conservation equations as defined by the FPV method. Example results of multidimensional problems including a turbulent Sandia jet and a reacting backward-facing step problem are presented for this method. Finally, conclusions of this approach are drawn with a description of future work.

MODEL FORMULATION

FLAMELET GENERATED MANIFOLDS

Solving individual, one-dimensional flames to populate the manifolds can be done either by recasting the reacting Navier-Stokes (RNS) equations in terms of conserved scalars, such as mixture fraction and enthalpy, or by solving the RNS equations in physical space before translating the solution to mixture fraction space. The latter approach is taken in this study and the Cantera Counterflow Diffusion Flame (CDF) model to solve the one-dimensional RNS equations is used to determine flame solutions as a function of strain rate [21]. The Cantera CDF model solves the governing equations of a one-dimensional steady-state axisymmetric stagnation flow given as,

$$\frac{\partial \rho u}{\partial z} + 2\rho V = 0 \tag{1a}$$

$$\rho u \frac{\partial Y_i}{\partial z} = -\frac{\partial j_i}{\partial z} + MW_i \dot{\omega}_i \tag{1b}$$

$$\rho u \frac{\partial V}{\partial z} + \rho V^2 = -\Lambda + \frac{\partial}{\partial z}\left(\mu \frac{\partial V}{\partial z}\right) \tag{1c}$$

$$\rho c_p u \frac{\partial T}{\partial z} = \frac{\partial}{\partial z}\left(\lambda \frac{\partial T}{\partial z}\right) - \sum_i j_i c_{p,i} \frac{\partial T}{\partial z} - \sum_i h_i MW_i \dot{\omega}_i \tag{1d}$$

where, ρ is the density, u is the axial velocity, $V = v/r$ is the scaled radial velocity, Y_i is the mass faction of the i^{th} species, j_i is the diffusive mass flux of the i^{th} species, Λ is the pressure eigenvalue (indpendent of z), T is the temperature, λ is the thermal conductivity, μ is the dynamic viscosity, $c_{p,i}$ is the constant pressure heat capacity, h_i is the enthalpy, MW_i is the molecular weight, and $\dot{\omega}_i$ is the molar production weight, all of the i^{th} species. The governing equations are computed along the stagnation streamline ($r = 0$), using a similarity solution to reduce the dimensionality of the problem to a $1D$ steady-state solution. See references [21] and [22] for more details.

The boundary conditions for the $1D$ flame solution are dependent on the intended use of the FGM. Fuel and oxidizer concentrations as well as their respective temperatures are Dirichlet boundary conditions explicitly set in the flame solution. As an example, for a reacting methane jet in atmospheric conditions, the fuel (CH_4) mass fraction is set to $Y_{CH4} = 1$ and the temperature is $T_f = 300\,K$, while the oxidizer (as air) is set to $Y_{O2} = 0.23$ and $Y_{N2} = 0.77$ and $T_{ox} = 300\,K$. The pressure is set to $P = 101325\,Pa$ and is constant between the $1D$ flame simulations and the large scale simulations for which the FGM is to be applied. The initial solution is obtained with as little strain rate as possible and then incrementally increased until the strain is so high, the reaction rates cannot keep up with the rate of diffusion of oxidizer and fuel into the flame zone and the flame extinguishes. Using the mixture fraction definition shown as,

$$Z_C = \sum_{i=1}^{n} \frac{a_{iC} MW_C}{MW_i} Y_i \tag{2}$$

where, Z_C is the mixture fraction as defined for species containing carbon atoms, a_{iC} is the number of carbon atoms within a molecule - e.g. for CH_4 $a_{iC} = 1$, MW is the molecular weight of carbon with subscript C or of the i^{th} species, and Y_i is the mass fraction. This definition of the mixture fraction illustrates its conserved scalar property by recognizing it as the definition of the distributed carbon atoms within a system where atoms must be conserved. Lastly, the mixture fraction can be defined with any molecule contained in the fuel species, as mixture fraction must very from $Z = 1$ at the fuel to $Z = 0$ at the oxidizer, and will have the same exact distribution if Z_H is used provided the diffusivities of species are assumed equal.

Mapping the solutions to mixture fraction space, we can show some representative results for a methane-air system and describe the manifold generation process. Figure 1 shows representative temperature profiles as a function of Z for a subset of CH_4-Air flames at ambient conditions. As the strain rate is increased the temperature decreases from its adiabatic flame temperature to nearly $400\,K$ lower before extinguishment occurs. The left-hand side of the plot is the air ($Z = 0$) and the right-hand side is the fuel ($Z = 1$) with the adiabatic and peak flame temperatures occurring at the stoichiometric ratio of fuel/air, defined as Z_{st}. For methane, the stoichiometric mixture fraction is defined as, $Z_{st} = Y_{O2}/(\nu Y_{CH4} + Y_{O2}) = 0.0544$. In total, 223 flames were captured between the fully diffusion controlled flame through the region where it becomes kinetically controlled and eventually extinguishes.

Now that we have a clear definition of the level of mixing between fuel and oxidizer, as well as the tentative location of the stoichiometric combustion, we must now determine a way to describe the extent of reaction from pure mixing to full reacted products. This is accomplished through the definition of the progress variable and stoichiometric progress variable (Λ), defined as

$$C_{pv} = \frac{\sum_{i=1}^{n} \alpha_i Y_i}{\sum_{i=1}^{n} \alpha_i Y_{i,st,eq}} \tag{3a}$$

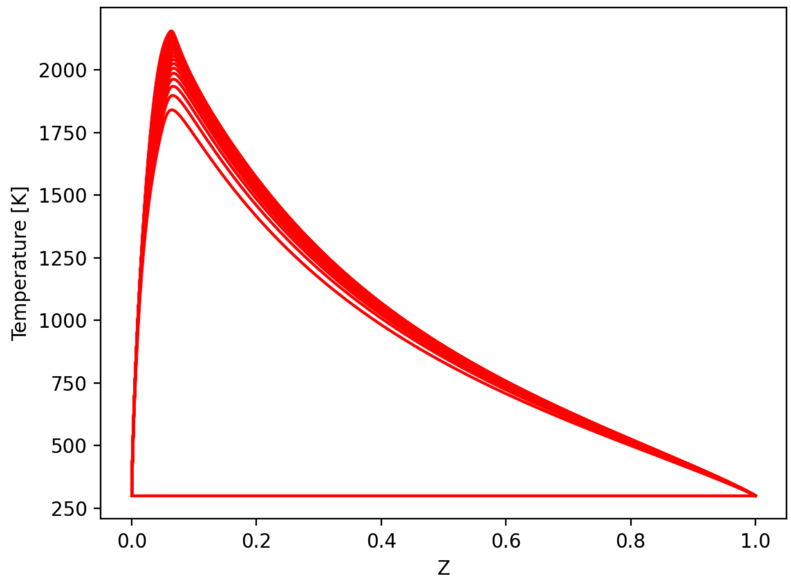
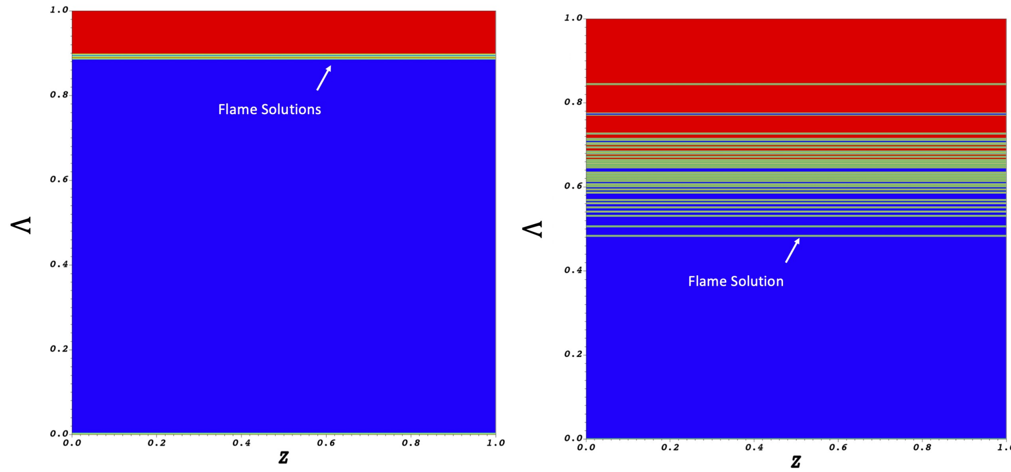


Figure 1: Representative temperature profiles vs. mixture fraction for a subset of methane-air flame solutions with increasing strain rate as the temperature decreases until extinguishment.

$$\Lambda = C_{pv,st} = \frac{\sum_{i=1}^{n} \alpha_i Y_{i,st}}{\sum_{i=1}^{n} \alpha_i Y_{i,st,eq}} \tag{3b}$$

where α_i are the weights (to be determined), and the subscripts st and eq represent the quantities at the stoichiometric and equilibrium (fully diffusion controlled solution at the lowest strain rate), respectively. Defining Λ in this way allows us to catalog an entire flame solution at a singular value of Λ, allowing us a statistically independent variable of Z that we can now use to determine the weights with a rigorous mathematical definition.

In this work we define a Heaviside cost function that contains the constraints for determining the α_i's coupled with a multivariate downhill simplex method to march the values towards a minimum. Previous studies have often used linear combinations of CO_2, H_2O, H_2, and CO [9, 23, 24]. In this study, we determine the starting species to define the progress variable by first eliminating them if they are found on either boundary, such that C_{pv} will vary between 0 and 1 as it progresses from unburnt to a completely combusted solution. Then, initial gradients of the species with respect to Z are taken to sort species to the bottom of the list by their values of dY/dZ - highest values go towards the bottom. This ensures the species used to define C_{pv} do not behave erratically during the combustion process, otherwise it could lead to pour results during lookup. Species are also sorted by their relevant maximum, favoring combustion products that peak near the lowest strain rate cases and vary monotonically as strain is increased. Finally, in order to keep the downhill simplex method efficient, a reasonable total number of species to define C_{pv} should be chosen - in our case, we decided to define C_{pv} with a total of four species. The species used for the methane-air combustion case, after sorting based on local maximums, their proximity to the boundary, and their gradients, were CO, H_2O, CH_3, and H_2 in descending order.



(a) (b)

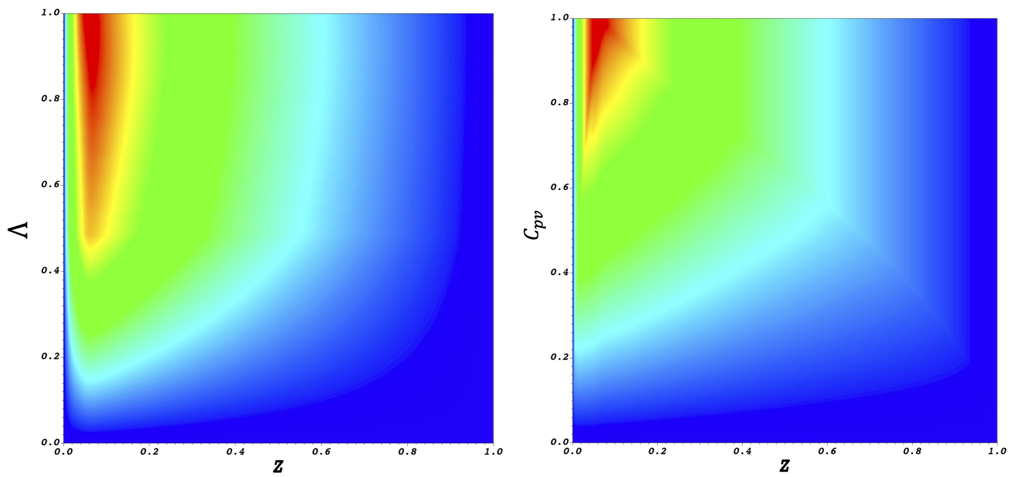
Figure 2: Difference is flame distribution (red lines) with simple weights (a) $\alpha_{CO} = 1$ and $\alpha_{H_2O} = 1$ versus the (b) rigorous definition with the downhill simplex method $\alpha_{CO} = 97$, $\alpha_{H_2O} = -56$, $\alpha_{CH_3} = 158$, and $\alpha_{H_2} = -7$.

Given our list of species, as determined simply by their properties in mixture fraction space, we can now use them and the definition of Λ to find the weights. To compute α_i, we employ and optimization strategy based on the work of Ihme *et al.* [24] and Bojko [17, 18], where a cost function is minimized. The cost function is defined as,

$$\iint \left[\Omega_1 H \left(-\frac{\partial C_{pv}}{\partial \Lambda} \right)_1 + \Omega_2 H \left(-VS \right)_2 + \Omega_3 H \left(|C_{pv}| - 1 \right)_3 + \Omega_4 H \left(-C_{pv} \right)_4 + \Omega_5 H \left(-SF \right)_5 \right] dZ d\Lambda \quad (4)$$

where, Ω is a penalty factor that can be weighted independently to give each term more credence when determining α_i, H is the Heaviside function, VS defines a valid solution in the bin-averaged discretized space (0 no solution - 1 is a valid solution), and SF is the sorted flame list to ensure flames stay in the same order as the low-to-high strain ordering. The Ω term in our case are all set to 1, but can be useful when certain criteria, such as flame ordering is not being met for a particular flame setup. The first Heaviside function enforces monotonicity of C_{pv} in Λ space so there is a single independent flame solution for every value of C_{pv}. The second Heaviside function encourages the weighted values to span as much of the Λ-Z space possible, which will reduce reliance on interpolation and reducing manifold gradients of tabulated species and source terms in the mapped solution. The third Heaviside function penalizes the solver for values of C_{pv} greater than 1, while the fourth term discouraging negative values of C_{pv}. The fifth Heaviside function penalizes the solution if the flames are out of order, i.e. if they no longer represent the low strain solution at $C_{pv} = 1$ and high strain extinguished solution at $C_{pv} = 0$. Taken together, all Heaviside functions mathematically define the need for a monotonic solution, with low gradients, with a C_{pv} that varies between 0 and 1, and maintains the physical relevance to the flame solution. The numerical solution procedure to minimize the cost function is a multidimensional downhill simplex method [25].

Figure 2 shows the difference between choosing simple weights (Fig. 2(a)), $\alpha_{CO} = 1$ and $\alpha_{H_2O} = 1$,



(a) (b)

Figure 3: Example of the temperature color map contour as plotted in (a) Λ-Z space and then (b) mapped to C_{pv}-Z space.

versus using the cost function analysis with the downhill simplex method to determine the weights (Fig. 2(b)) as, $\alpha_{CO} = 96.07$, $\alpha_{H_2O} = -56.19$, $\alpha_{CH_3} = 157.83$, and $\alpha_{H_2} = -7.35$. Specifically this highlights the distribution of the flame solutions and demonstrates how the accuracy of differentiating between the solution variables during lookup would require an increase in the number of nodes required to discretize the Λ-Z space, otherwise multiple solutions may either be bin averaged or completely discarded. Furthermore, the gradients in Λ and eventually C_{pv} will be much higher using a simple method. Considering the tabulation methods are known to suffer from the overuse of available storage and memory, where tables can be on the order of gigabytes, it is not advantageous to force solutions onto ever increasing domain sizes in order to maintain the numerical accuracy.

The final step in creating the flamelet generated manifolds, is to translate the solution from Λ-Z space to C_{pv} and Z space for direct numerical simulations or also including mixture fraction variance (Z_v) and an assumed probability distribution function (PDF) for Z and C_{pv}. This approach has been demonstrated previously, where a beta PDF and Dirac delta function are used to define the probability distributions of Z and C_{pv} respectively, with analytical descriptions of the independent variable distributions as a function of the perturbed dependent variables. This study assumes a Dirac delta function probability distribution for Z and C_{pv} for turbulent situations where we can not directly resolve the transport of these species at the subgrid scale, otherwise we are solving a direct numerical simulation with appropriate grid spacing to accurately capture the distribution of Z and C_{pv}. Figure 3(a) shows a representation of the temperature color map contour after interpolation between values of Λ and Z where no solution exists. One can imagine taking a slice at a constant Λ and obtaining the exact temperature profile for the flame versus Z. It can be seen then from this perspective that the flame temperature is distributed around Z_{st} and diminishes with decreasing Λ as we would expect. This idealization is lost when we convert the solution to C_{pv} and Z as shown in Fig. 3 because C_pv is not constant in Z, as we described earlier when determining viable species to not be located on the boundary and have low gradients in Z. The outline of the equilibrium solution ($\Lambda = 1$) can be seen as the faint line that increases from $Z = 0$ and $C_{pv} = 0$ up to $C_{pv} = 1$ at $Z = Z_{st}$ and then back down to 0 as $Z \to 1$. The color gradient along the top of this curve extends to $C_{pv} = 1$ to ensure a valid solution exists in this space, which is assumed to be the equilibrium solution.

FLAMELET PROGRESS VARIABLE APPROACH

To define the flamelet progress variable (FPV) approach in the context of the compressible Navier-Stokes equations, we must first assume there to be equal diffusivities amongst all the species. The transport equation for species mass fractions is given as,

$$\frac{\partial \rho Y_i}{\partial t} + \nabla(\rho \boldsymbol{v} Y_i) = \nabla \cdot \left(\rho D \frac{\partial Y_i}{\partial \boldsymbol{x}}\right) + \dot{w}_i \tag{5}$$

where $\boldsymbol{v}$ is the velocity vector, and D is the diffusion coefficient. Assuming D is constant allows for the definition of as defined in Eq. , to be used to create a linear summation of all species equations to obtain the transport equation for the mixture fraction. This results in a cancellation of the source terms $\dot{w}_i$ because any creation or destruction of species containing elemental carbon will balance, leading to the conserved scalar transport equation. Similarly, the definition of C_{pv} can be created by multiplying and dividing through by the weights and constant values at the equilibrium solution, after summation becomes the transport equation for the progress variable. In this instance, C_{pv} is not a conserved scalar and will contain the species weighted source term $\dot{S}'''_{Cpv}$ to describe the creation and destruction of the progress variable in relation to the $1D$ flame solution.

The compressible Navier-Stokes equations with the FPV applied can be represented as,

$$\frac{\partial \boldsymbol{y}}{\partial t} + \nabla \cdot (F^c(\boldsymbol{y} - F^v(\boldsymbol{y}, \nabla \boldsymbol{y})) - \dot{S}(\boldsymbol{y}) = 0 \tag{6a}$$

$$\boldsymbol{y} = (\rho, \rho Z, \rho C_{pv}, \rho \boldsymbol{v}, \rho E) \tag{6b}$$

$$F^c(\boldsymbol{y}) = (\rho \boldsymbol{v}, \rho Z \boldsymbol{v}, \rho C_{pv} \boldsymbol{v}, \rho \boldsymbol{v} \cdot \boldsymbol{v} + pI, (\rho E + p)\boldsymbol{v}) \tag{6c}$$

$$F^v(\boldsymbol{y}) = \left(0, \rho D Z, \rho D C_{pv}, \tau, k \nabla T + \boldsymbol{v} \cdot \tau - \sum_i \rho D h_i \nabla Y_i\right) \tag{6d}$$

$$\dot{S}(\boldsymbol{y}) = \left(0, 0, \dot{S}'''_{Cpv}, 0, 0\right) \tag{6e}$$

where $\boldsymbol{y}$ is the fluid state vector including the conservative variables of mass, mixture fraction, progress variable, momentum, and energy. $F^c(\boldsymbol{y})$ is the convective flux terms and $F^v(\boldsymbol{y})$ is the viscous flux terms. Lastly, $\dot{S}(\boldsymbol{y})$ is the volumetric source term vector which only contains the source term for the progress variable. All thermodynamic variables, i.e. h_i, c_p, k, etc., are determined from the species from the table, $Y_i = f(Z, C_{pv})$, and the mass weight thermodynamic fits. Species gradients, ∇Y_i are determined using the chain rule such that, $\nabla Y_i = dZ/d\boldsymbol{x} \cdot dY/dZ + dC_{pv}/d\boldsymbol{x} \cdot dY/dC_{pv}$. The gradients of mixture fraction and progress variable can be determined directly from the flow solver and gradients of Y_i with respect to flamelet variables can be derived from the table. Unlike other flamelet formulations, the above retains the fully conservative and compressible form of the NS equations, with the caveat that reaction rates are calculated for a singular pressure. Aside from the reaction rates, the chemical makeup of the flame solutions is assumed not to change considerably with pressure, given the solution at pure mixing or at equilibrium is the same regardless of the given pressure. Equations 6a-6e are solved in the JENRE® Multiphysics framework using a discontinuous galerkin (DG) approach which has the advantages of a high order of

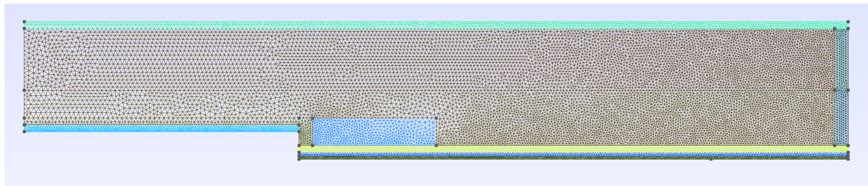


Figure 4: Two-dimensional grid showing the backward facing step, flow is moving left to right with the smallest physical spacing being $dx = 0.4\,mm$ near the walls and largest value being $dx = 2\,mm$ near the centerline.

accuracy on unstructured grids.

RESULTS AND DISCUSSION

To test the FPV approach, the NASA data for a backward facing step flow is used to validate the flow solver [26] with a step height of $h = 9.8\,mm$ to create a computation domain with enough mesh refinement for implicit large eddy simulations (iLES). To start, the overall length of the domain is $L = 3h$, the height of the domain is $H = 5h$, and the width of the domain (for 3D runs) is $W = 4h$ with periodic boundary conditions. The inlet length and height are set to $L_i = 10h$ and $H_i = 4h$, respectively. The computational domain includes grid compression at the walls and in the recirculation zones at ratios of $dx/5$ and $dx/4$, while 2D studies will be used to determine the appropriate size of dx. Initial testing of dx sizing will occur from $dx = 1$ to $5\,mm$ on a 2D grid to observe the effects on the recirculation zone and reattachment length. The initial Reynolds number will be set for to $Re = 5000$, but additional Re numbers will be explored, then we will add CH_4 blowing in from the bottom wall without reactions, then finally ignite the mixture and explore the effects of combustion on the recirculation zone. With the use of the DG method, we not only have the ability to refine the grid sizing, i.e. dx, but we can also define the polynomial order p used to represent the solution on each physical element - we chose $p = 2$ such that the solution can be considered to have a representative dx of at least half of the values we are showing. That is, our dx values are split further within the DG representation than the physical grid represents. Figure 4 shows the representative $2D$ grid we are solving the backward facing step flow problem with a $dx = 2\,mm$ at its largest values and $dx = 0.4\,mm$ near the walls. The problem of considering a diffusion flame with a backward facing step is relatively unexplored, while the combustion problem of a premixed flame with the backward facing step has explored by Pouech et al. [27].

The operating fluid for a no reacting case is air at $T = 300\,K$ and $P = 1\,atm$ and with a density of, $\rho = 1.17197034943965 47\,kg/m^3$ and the dynamic viscosity is $\mu = 1.8629857568444622e^{-5}\,Pa \cdot s$, so that with a $Re = 5000$. the velocity at the inflow is $u_x = 8.1102985\,m/s$. These values are used to set the subsonic characteristic inflow condition at the left-hand side. The top boundary condition is set as a slip wall, the bottom and side walls are adiabatic slip walls, and the outflow is a subsonic characteristic outflow. When activated, the bottom fuel boundary ranges from just below the step to $15h$ downstream, such that it does not span the entirety of the domain. The fuel in this case is methane CH_4 and will blow from the wall with a set velocity of $u_y = 0.9173\,m/s$ at a temperature of $T_f = 300\,K$ and with a density of $\rho_f = 0.6517\,kg/m^3$. Figure 5 shows the velocity color map for a maximum velocity of $14\,m/s$ (red) and a minimum velocity of $-5\,m/s$ (blue) for the case with no fuel blowing (top) and with fuel blowing (bottom). The recirculation zone for the case without fuel blowing tends to be slightly shorter on average, but both cases tend to have

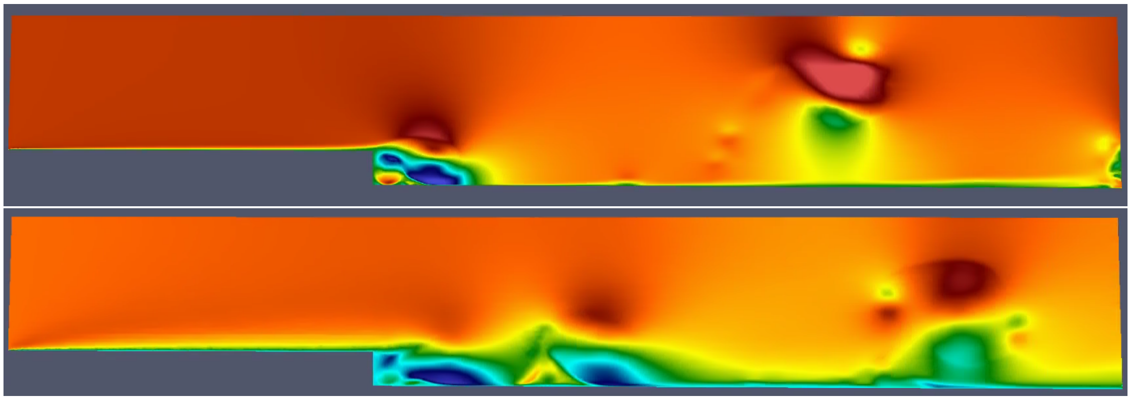


Figure 5: Velocity (v_x) contours with a maximum of $14\,m/s$ (red) and minimum of $-5\,m/s$ (blue) for the case with no fuel flow at the bottom boundary (top) and with fuel flow (bottom), both images are for $t = 0.055\,s$.

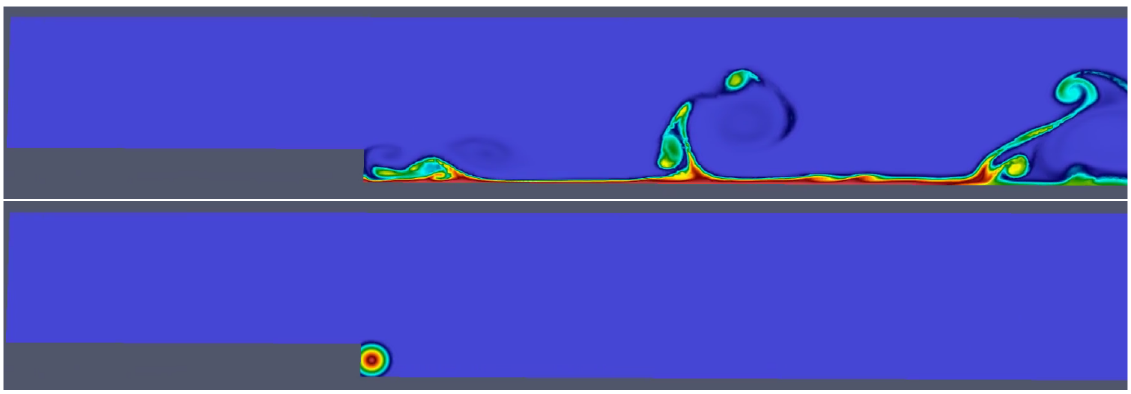


Figure 6: Color contours of the (a) mixture fraction ranging between $Z = 0$ (blue) and $Z = 1$ (red) and (b) the ignition kernel with $C_{pv} = 1$ (red) and $C_{pv} = 0$ (blue).

a similar vortex shedding frequency of approximately one every $0.02\,s$. These vortices can be observed moving through the domain in the right of both figures with a dark red top and bottom green portion of the vortex, indicating a positive flow at the top and negative flow at the bottom as it rotates through the domain.

In order to ignite the mixture building up behind the step, we have to seed the flow with a positive value for C_{pv} above the threshold for which extinguishment will occur. Figure 6(a) shows the mixture fraction distribution after $t = 0.05\,s$ into the solution time, at this point there have been a couple of shedded vortices to come through the fuel zone and lift it into the free stream to encourage mixing. Just after this instant in time, an ignition kernel is placed just behind the step in the form of a radial distribution of the progress variable. In this case, the solution is read into a post-processing algorithm, the value of C_{pv} is set and the temperature and density are updated based on the current value of Z and P within the region, these values are then used to determine and set the state ($\boldsymbol{y}$) before the simulation is restarted.

Figure 7 shows the conditions of the velocity (top), mixture fraction (middle), and the temperature (bottom) after ignition has taken place and the flame has had a chance to surpass its initial transients and reach a steadier solution. The same ranges of the plots are kept the same for sake of comparison, while

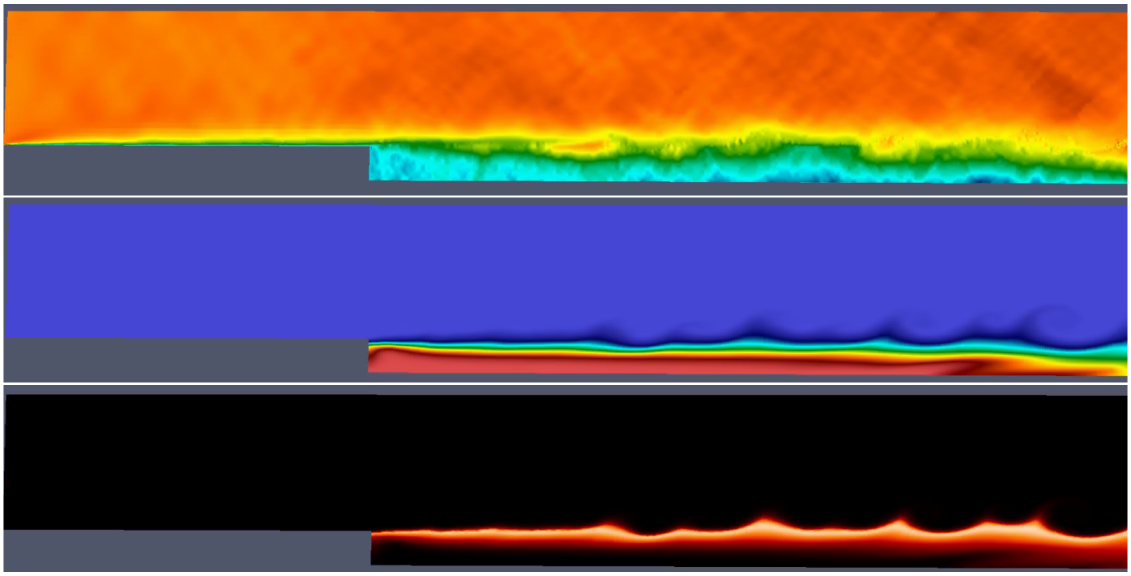


Figure 7: Color contours of the (top) the x-dir velocity ranging from -5 to $14\,m/s$ (blue-to-red), (middle) mixture fraction distribution ranging between $Z = 0$ (blue) and $Z = 1$ (red), and (bottom) temperature contours ranging from dark ($T = 300\,K$) to bright orange ($T = 1800\,K$).

the temperature contours range from the dark region at $T = 300\,K$ to the bright orange at $T = 1800\,K$. The first notable difference in the reacting cases is the difference in the velocity profiles, while the non-reacting cases demonstrate the typical recirculation zone we come to expect from the backward facing step problems, the reacting flow and heat release wash the recirculation zone out and no more shedding frequencies or clearly defined recirculation region can be identified. Furthermore, the velocity patterns tend to be much more dynamic as the local ignition kernels cause slight jumps in pressure that reflect on the upper and lower boundaries within the domaine, revealing intersecting wave patterns within the flow-field. The evidence of the recirculation zone is further evidenced by the mixture fraction distribution which now appears as a uniform bottom boundary to the shear flow occurring above. There is a noticeable mixing layer between the air, hot gases, and the fuel zone, but there is no longer a clear penetration and vortical mixing as was observed in the non-reacting cases. Finally, the temperature profile highlights the region of the highest reaction rates; again this is found at the shear layer between the air and fuel zone, with a subtle wave formation observed. It is also worth noting, the initial ignition process maintained combustion temperatures close to the adiabatic flame temperatures of methane and air, around $T = 2150\,K$. However, as the flame progressed to a steadier profile, the peak temperature settled to $T = 1850\,K$. This implies there is enough strain in the shear layer to keep the reactions from progressing to the equilibrium flame temperature solutions.

CONCLUSION

This study lays out the flamelet progress variable (FPV) approach with the use of flamelet generated manifolds (FGM) to model the detailed combustion of the compressible fully conservative reacting Navier-Stokes equations using a discontinuous galerkin (DG) flow solver with a polynomial degree of $p = 2$. Firstly, the process for creating and tabulating individual flame solutions is laid out with definitions of the mixture fraction (Z) and progress variable (C_{pv}) for translating the flame solutions to the flamelet variable space. The initial step requires a mapping to Λ-Z space in order to minimize the user defined cost function using the downhill simplex approach in a mathematically rigorous procedure in order to determine the weights (α's) used to define C_{pv}. This method was shown to substantially spread out the flame solutions in order to reduce gradients and increase the accuracy of the tabulated data. It was then demonstrated how to transform the solution from Λ-Z space to C_{pv}-Z space in order to use these tables in the FPV approach. The compressible and fully conservative form of the Navier-Stokes equations are expressed in terms of the mass, mixture fraction, progress variable, momentum, and energy conservation equations. This reduces the overall species partial differential equations to be solved from $N-1$, where N is the number of species, to two, while the numerical stiffness is significantly reduced by accounting for reaction progression with C_{pv}.

The resulting configuration was then demonstrated on a backward facing step problem with a step height of $9.8\,mm$ in order to compare to experimental data. Three different flow simulations were carried for a Reynolds number of $Re = 5000$ for the inflow condition, assuming air at ambient conditions as the operational fluid. Non-reacting flow showed the expected recirculation zone with a vortex shedding frequency of approximately $50\,Hz$. Similarly, when the bottom fuel boundary was turned on, with a flow velocity roughly 100 times smaller than the inflow, a similar shedding frequency was observed along with clear mixing layers between the fuel and air flow as the vortices carried fuel to the core flow as the traverse through the domain. A slightly longer recirculation zone was observed when fuel blowing was turned on versus the unmixed flow only. Lastly, the mixed boundary layer was ignited with a progress variable hot spot just behind the step region. Ignition was highly transient causing pressure waves and velocity disturbances as the fuel quickly converted to hot reactants. Eventually, the ignition process settled out and a steady solution was observed for the reacting case. The mixing between fuel and oxidizer was significantly hampered by the heat release zone and no more vortical shedding or thing mixing layers generated from the fuel surface are observed. The shear layer sets up at the fuel and oxidizer boundary, exactly where the heat release is occurring; however, it is occurring with enough strain to limit the overall adiabatic flame temperature from its peak value of $T_{ad} = 2150\,K$ to temperatures that peak near $T = 1850\,K$.

The FPV approach is a numerically efficient solver with the ability of including the effects of finite-rate chemistry for reacting flow simulations at large-scales. Work on the comparison to experimental data for the backward facing step is ongoing and continued validation of the FPV approach is underway. Additionally, turbulent flames such as those found in the International Workshop on Measurement and Computation of Turbulent Flames (TNF) database [28] are currently being simulated to validate the modeling approach.

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